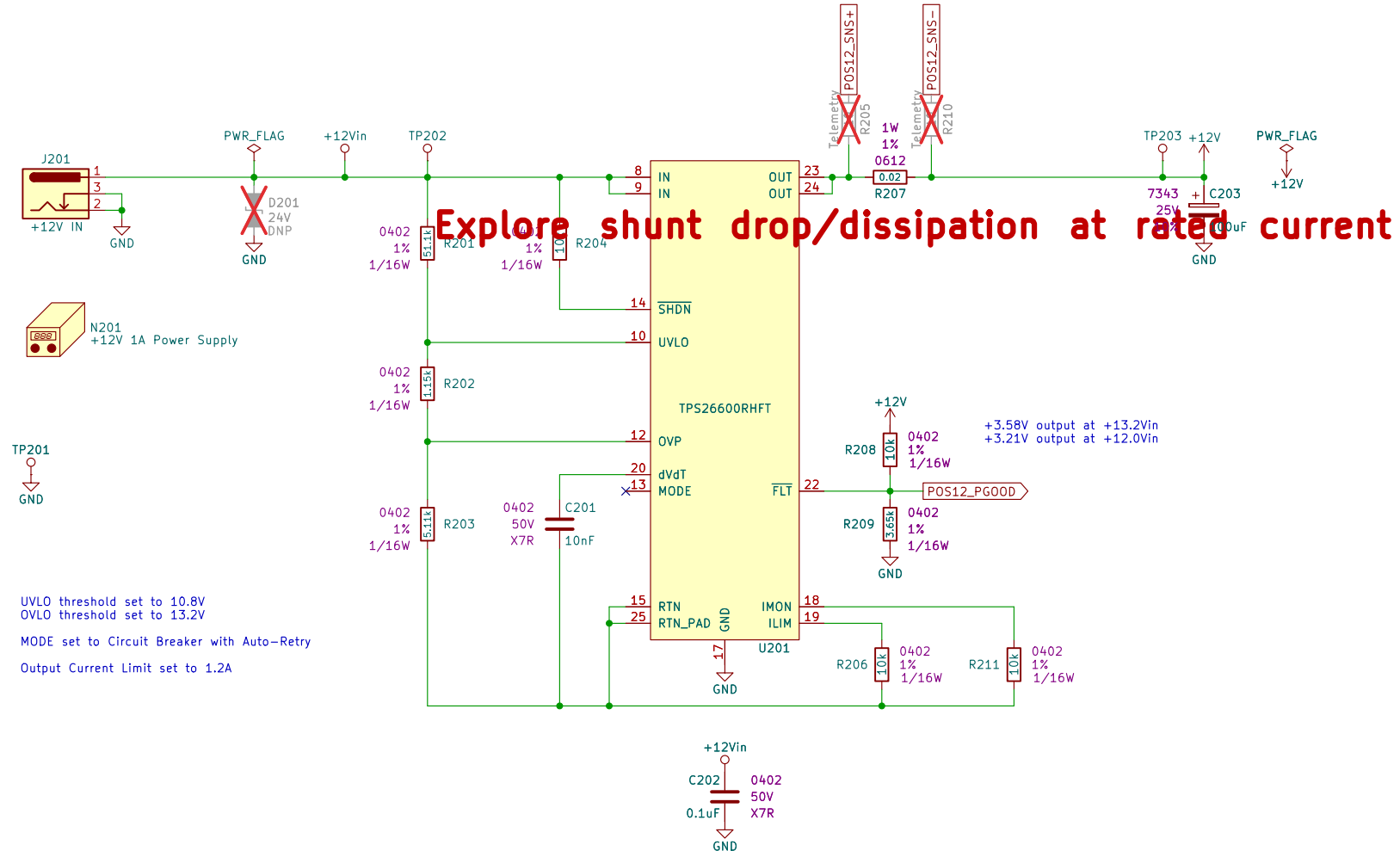
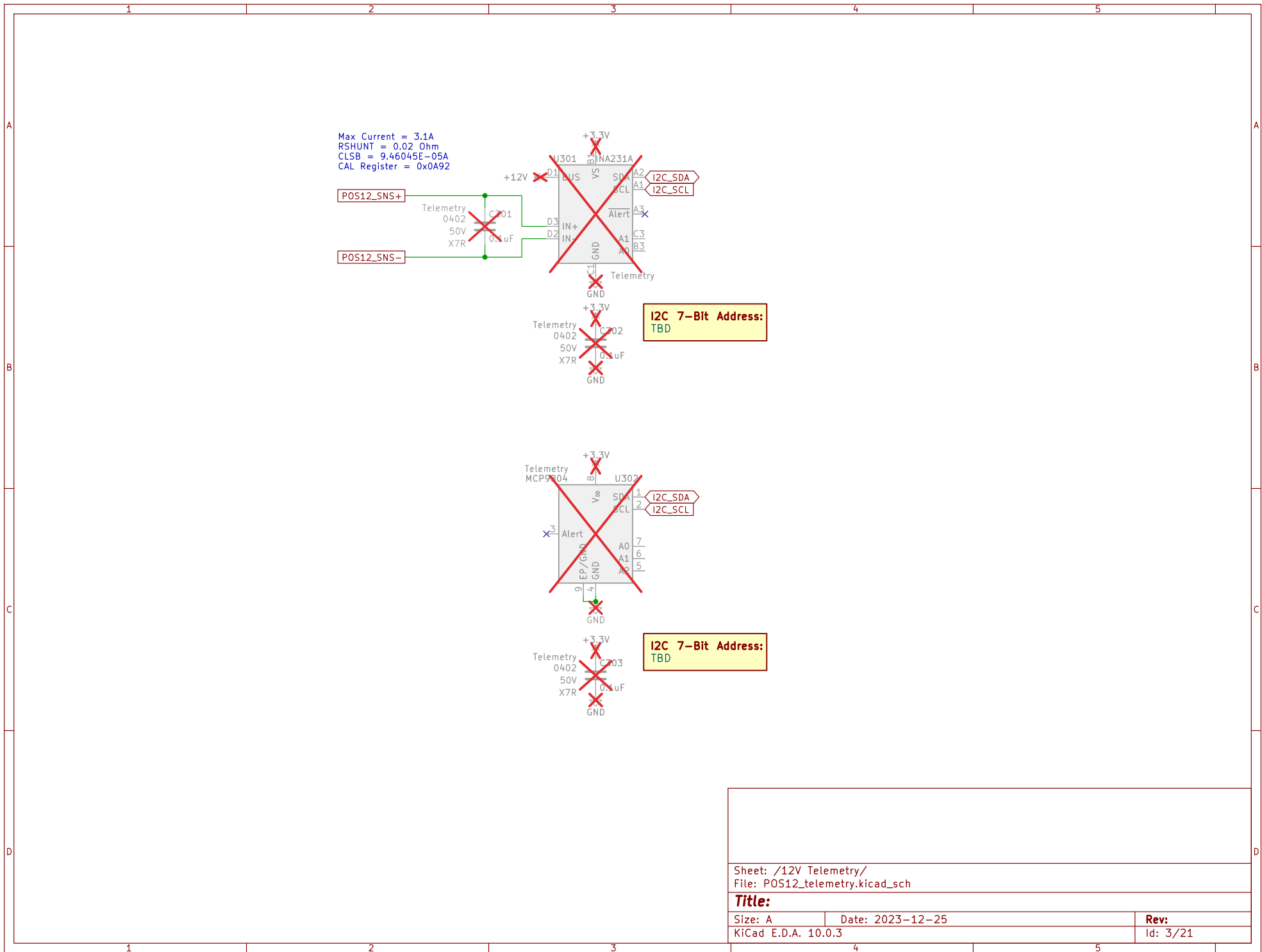


1	2	3	4	5	
A	12V Power Input File: POS12_power_input.kicad_sch 12V Telemetry File: POS12_telemetry.kicad_sch 3.3V Power Supply File: POS3P3_power_supply.kicad_sch 3.3V Telemetry File: POS3P3_telemetry.kicad_sch dsPIC33 Programming File: dspic33_programming.kicad_sch dsPIC33 Power File: dspic33_power.kicad_sch dsPIC33 GPIO File: dspic33_gpio.kicad_sch USB UART Bridge File: usb_uart_bridge.kicad_sch	Platform ETC File: platform_etc.kicad_sch PGOOD LEDs File: pgood_leds.kicad_sch Status LEDs File: status_leds.kicad_sch RCA Signal Conditioning File: rca_signal_conditioning.kicad_sch Speaker Signal Conditioning File: speaker_signal_conditoning.kicad_sch Source Selection, PGAs File: source_selection_pgas.kicad_sch Anti-alias Filtering File: anti_alias_filtering.kicad_sch Audio Voltage References File: audio_voltage_references.kicad_sch LED Panel Buffers File: led_panel_buffers.kicad_sch Pushbuttons File: pushbuttons.kicad_sch Mechanical File: mechanical.kicad_sch LED Panel Connectors File: led_panel_connectors.kicad_sch			
B					
C					
D					
1	2	3	4	5	

1.2A is not enough, this device will most likely need to be changed





1 2 3 4 5

A

B

C

D

1 2 3 4 5

D

SHRINK

1206 25V X5R C402 10uF

1206 25V X5R C403 10uF

0402 50V X7R C405 0.1uF

0402 1% 1/16W R402 10k

74LVC1G97 U401

POS3P3_PG00D

0402 1% 1/16W R401 10k

0402 50V X7R C401 0.1uF

POS3P3_RUN

0402 50V X7R C404 10nF

0402 1% 1/16W R409 1k

0402 1% 1/16W R410 1k

TPS62130A U402

PVIN AVIN

EN

PG

SS/TR

DEF

FSW

AGND EP PGND

SW

VOS

FB

20% 6.7A IHLP2020 2.2uH L401

0805 10V X7R C406 10uF

0805 10V X7R C407 10uF

0402 1% 1/16W R404 0.02

0402 1% 1/16W R405 1/16W

TP401

+3.3V PWR_FLAG

+3.3V

Telemetry POS3P3_SNS+

Telemetry POS3P3_SNS-

0402 1% 1/16W R403 0

POS12_PG00D

POS3P3_RUN

BAT54J D401

0805 10V X7R C408 10uF

0402 1% 1/16W R408 10k

Q401 BSS214NW

Q402 ZXMS6004FFTA

Load	Draw	Quantity	Total
dsPIC	50mA	1	50mA
8MHz dsPIC Clock	7mA	1	7mA
MCP6N11	1.1mA	4	4.4mA
MCP6024	5.4mA	2	10.8mA
Analog Switch	0.05mA	1	0.05mA
Digipot	0.6mA	1	0.6mA
1.6V Reference	0.2mA	2	0.4mA
PCA9536 GPIO Expander	0.29mA	1	0.29mA
I2C Buffers	8.3mA	3	24.9mA
Octal Buffers	1mA	2	2mA
LVC1G19s	0.01mA	21	0.21mA
Rotary Encoder	50mA	1	50mA
Module	1.3mA	11	14.3mA
INA231	0.42mA	2	0.84mA
MCP9804	0.4mA	3	1.2mA
ETC	0.36mA	1	0.36mA
Cap Touch Sensor	0.729mA	1	0.729mA
Total		0.554W	168mA

Sheet: /3.3V Power Supply/
File: POS3P3_power_supply.kicad_sch

Title:

Size: A Date: 2023-12-25 Rev:

KiCad E.D.A. 10.0.3 Id: 4/21

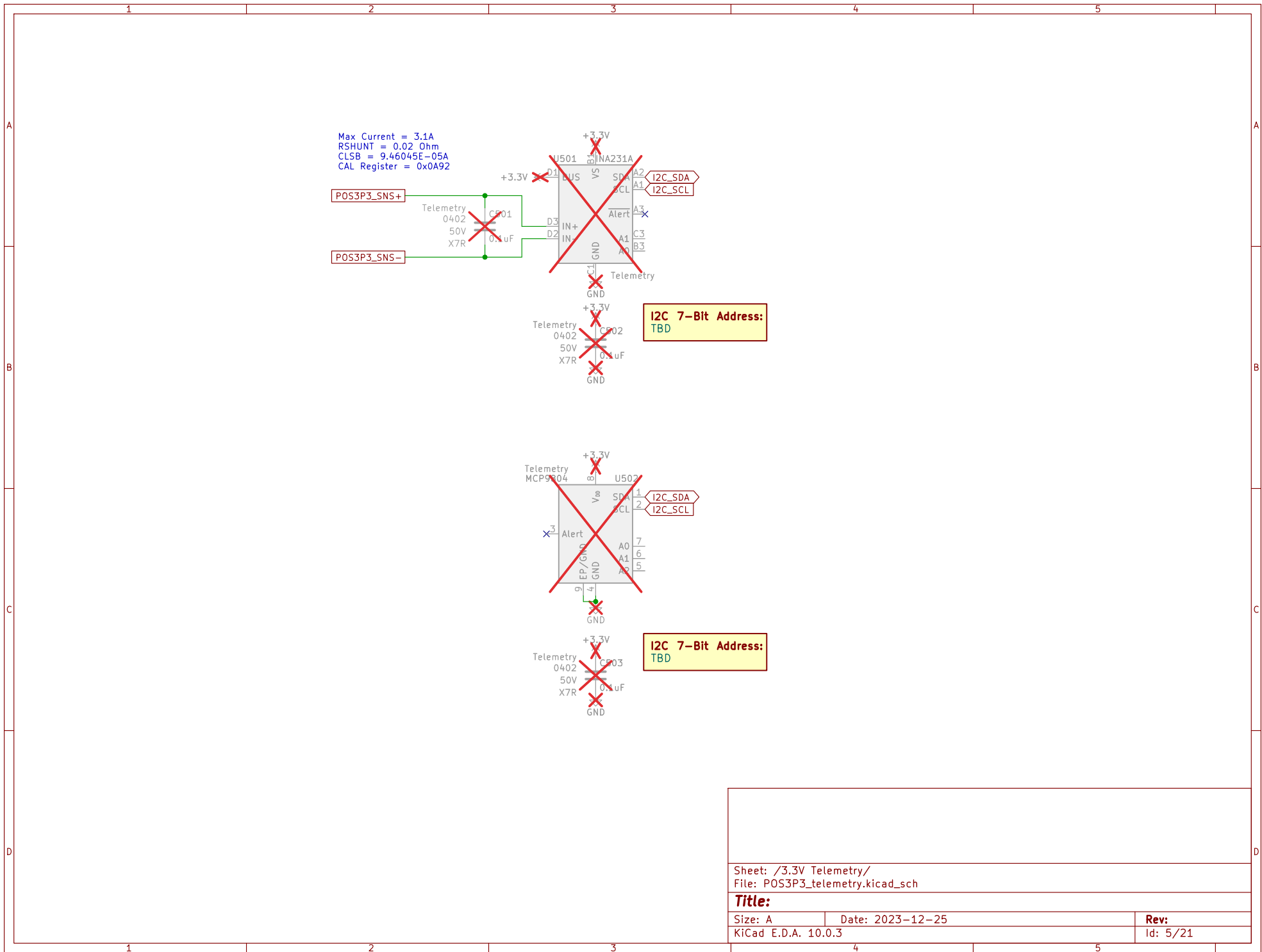
Load	Draw	Quantity	Total
dsPIC	50mA	1	50mA
8MHz dsPIC Clock	7mA	1	7mA
MCP6N11	1.1mA	4	4.4mA
MCP6024	5.4mA	2	10.8mA
Analog_Switch	0.05mA	1	0.05mA
DigiPot	0.6mA	1	0.6mA
1.6V_Reference	0.2mA	2	0.4mA
PCA9536 GPIO Expander	0.29mA	1	0.29mA
I2C_Buffers	8.3mA	3	24.9mA
Octal_Buffers	1mA	2	2mA
LVC1G19s	0.01mA	21	0.21mA
Rotary Encoder Module	50mA	1	50mA
INA231	1.3mA	11	14.3mA
INA231	0.42mA	2	0.84mA
MCP9804	0.4mA	3	1.2mA
ETC	0.36mA	1	0.36mA
Cap Touch Sensor	0.729mA	1	0.729mA
Total		0.554W	168mA

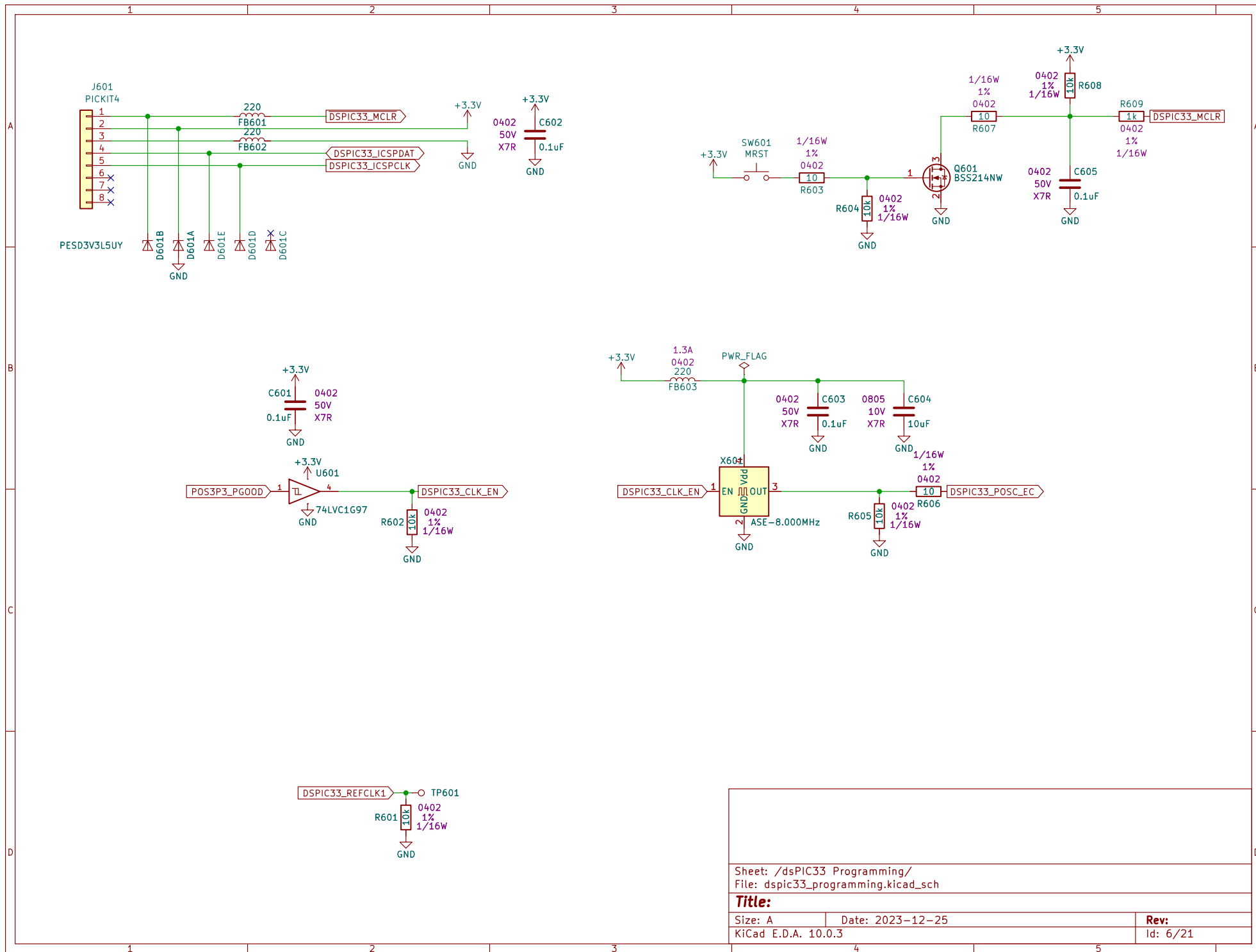
Sheet: /3.3V Power Supply/
File: POS3P3_power_supply.kicad_sch

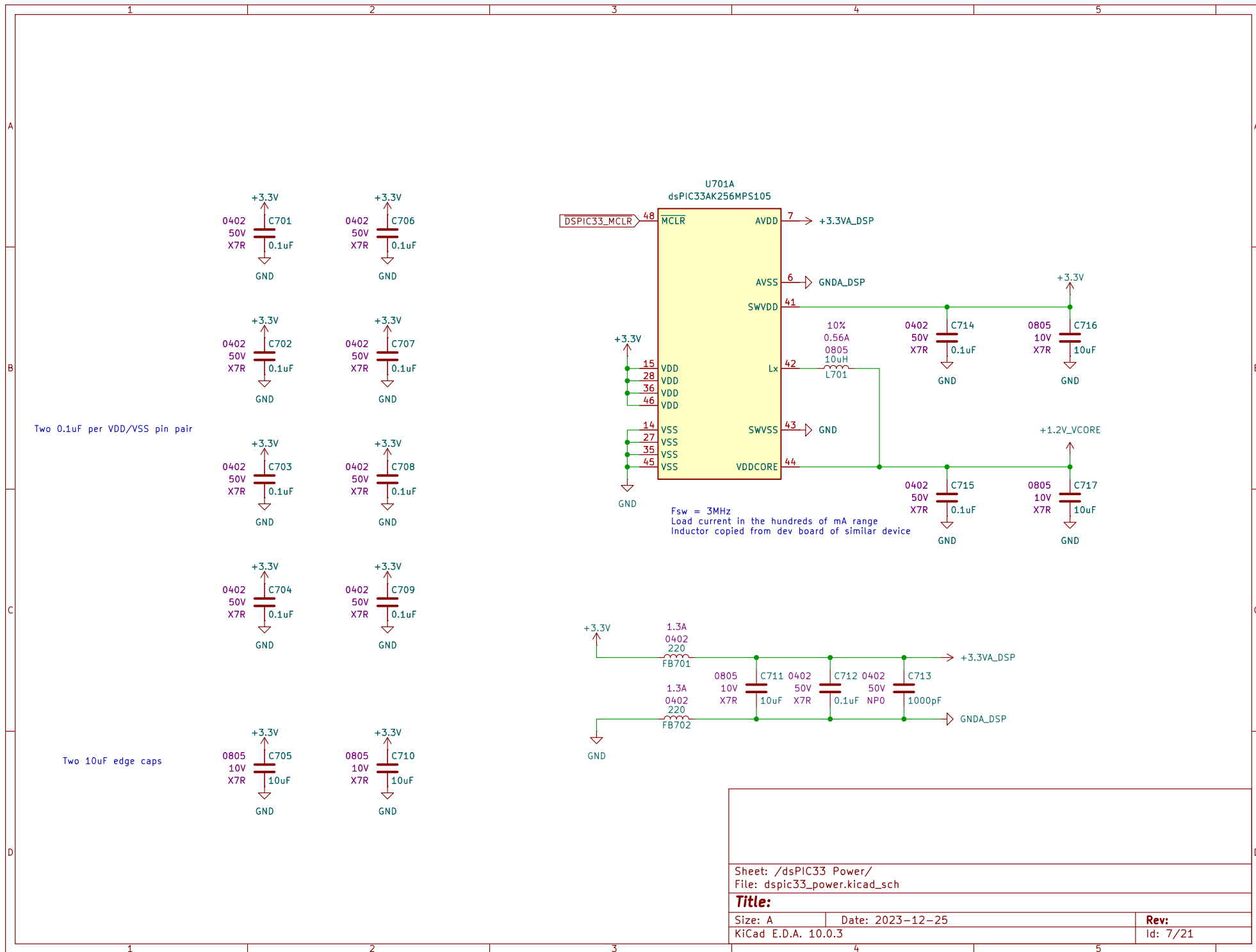
Title:

Size: A	Date: 2023-12-25
KiCad E.D.A. 10.0.3	

Rev:
Id: 4/21





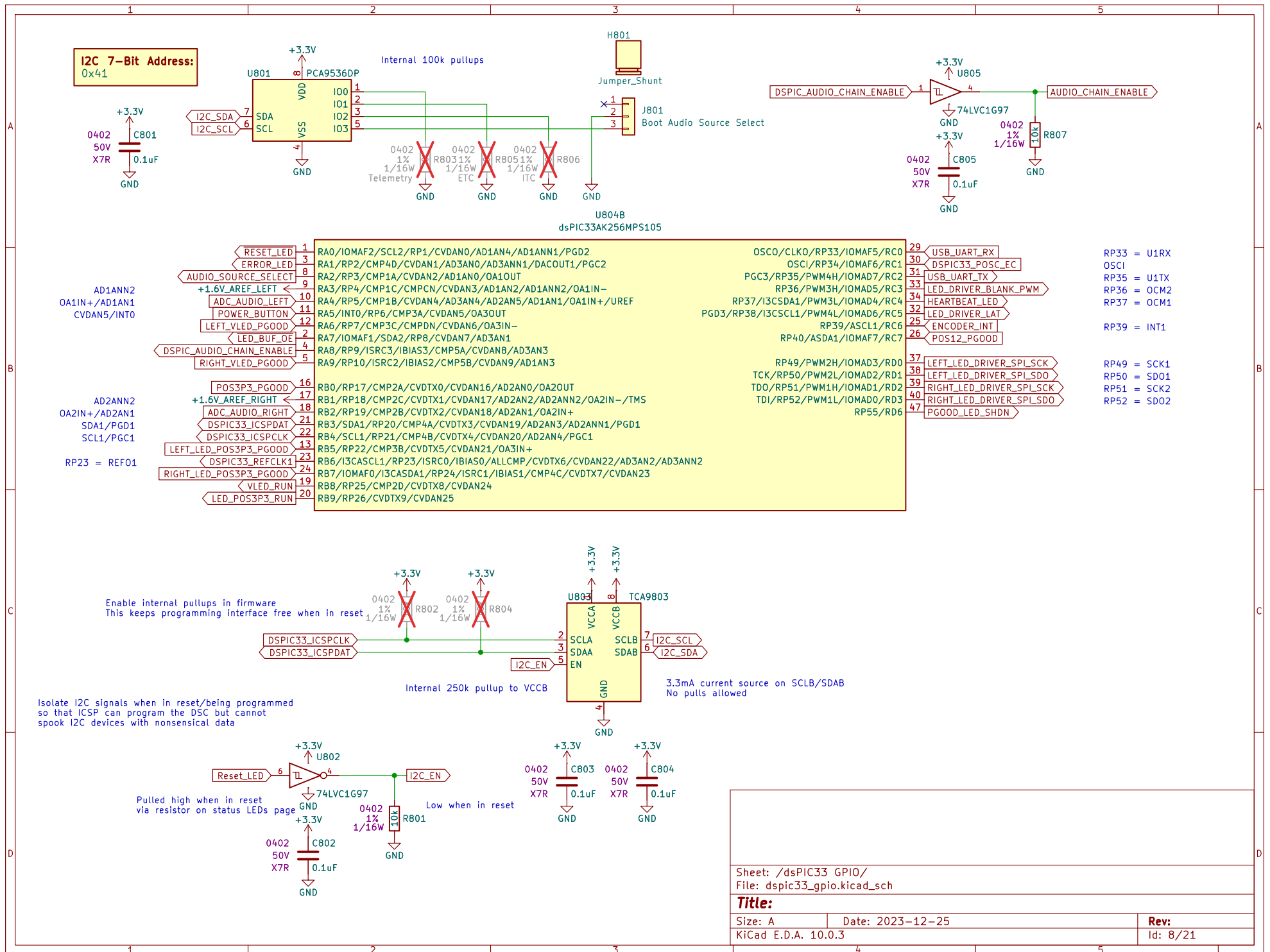


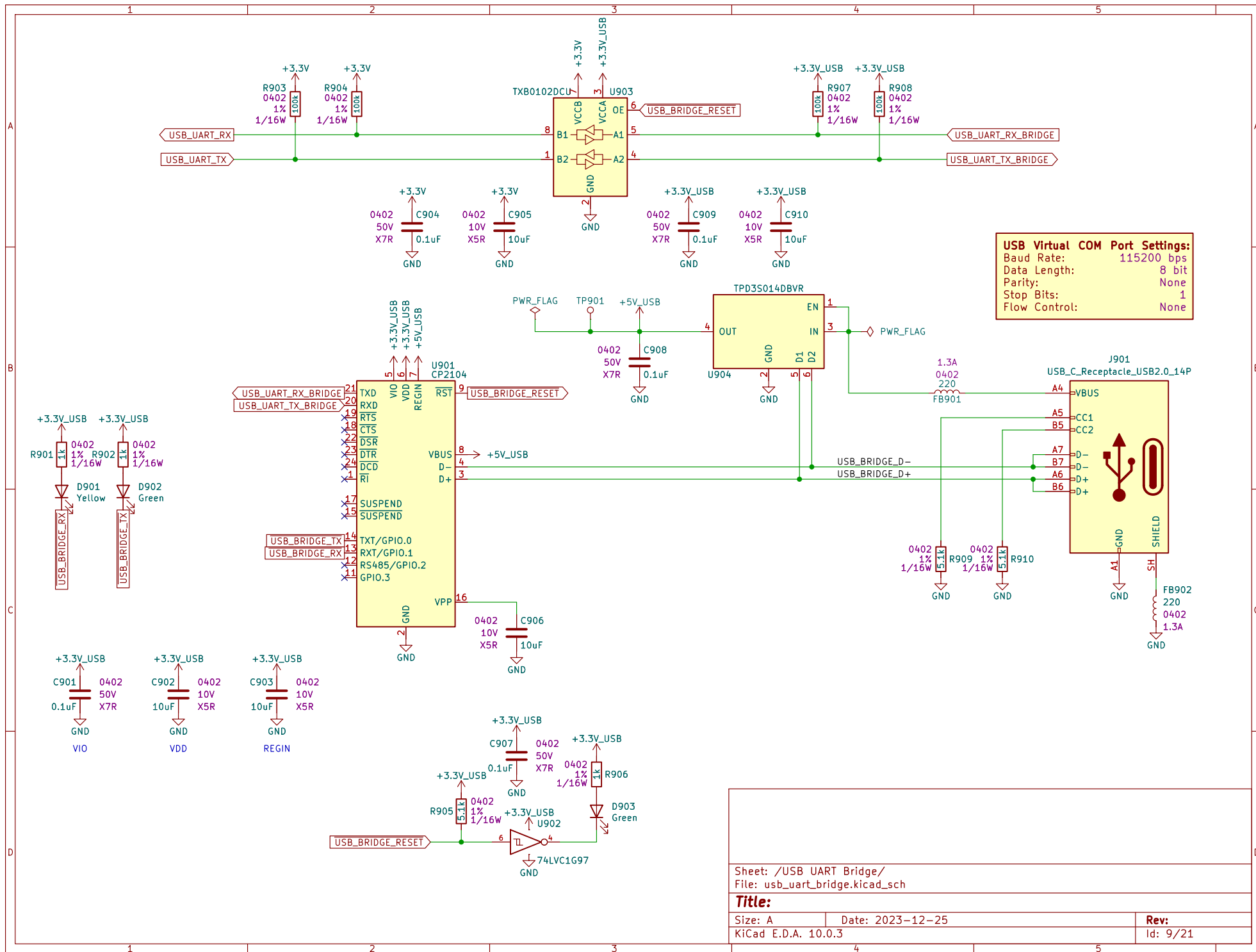
Sheet: /dsPIC33 Power/
File: dspic33_power.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 7/21



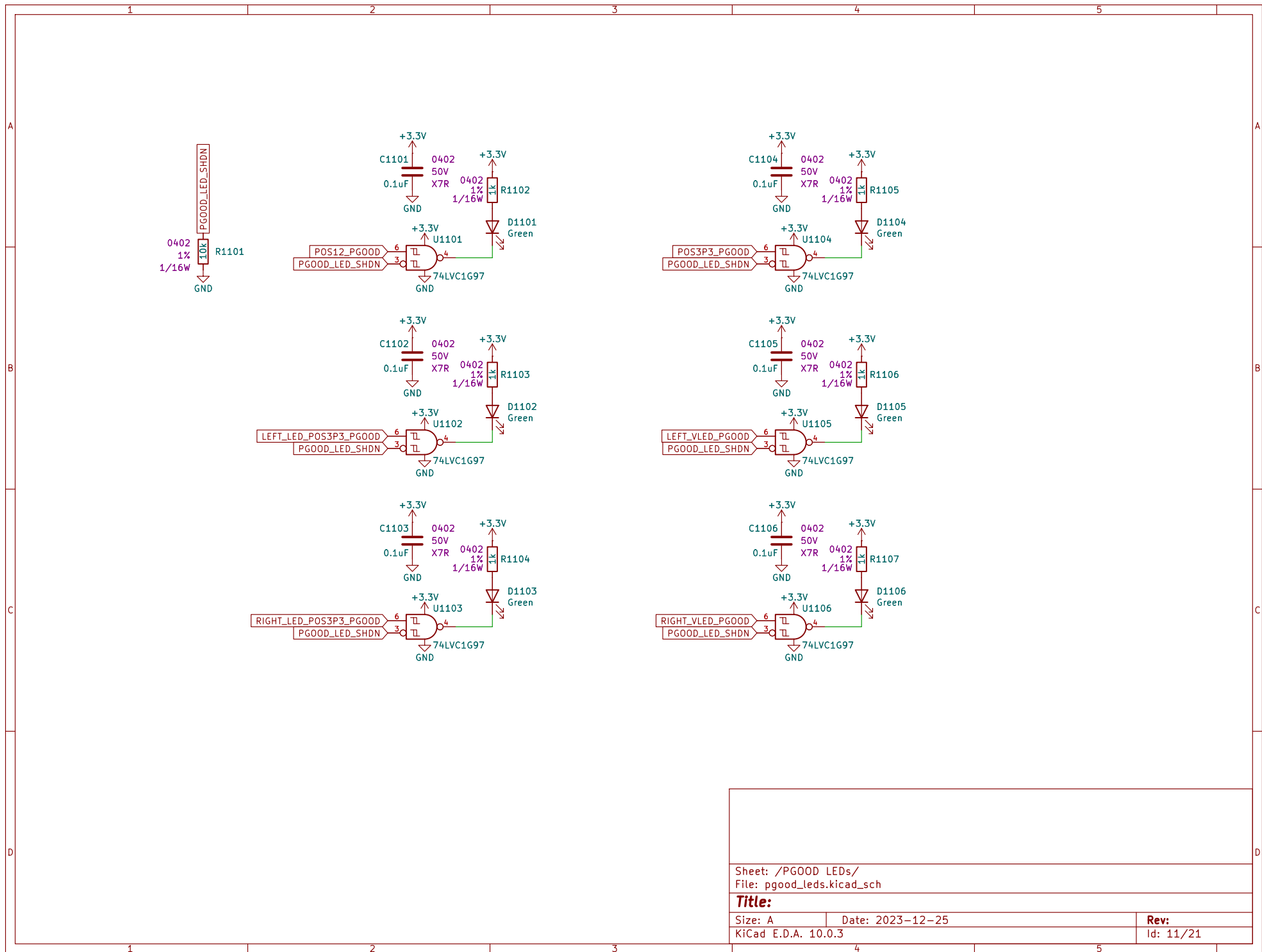


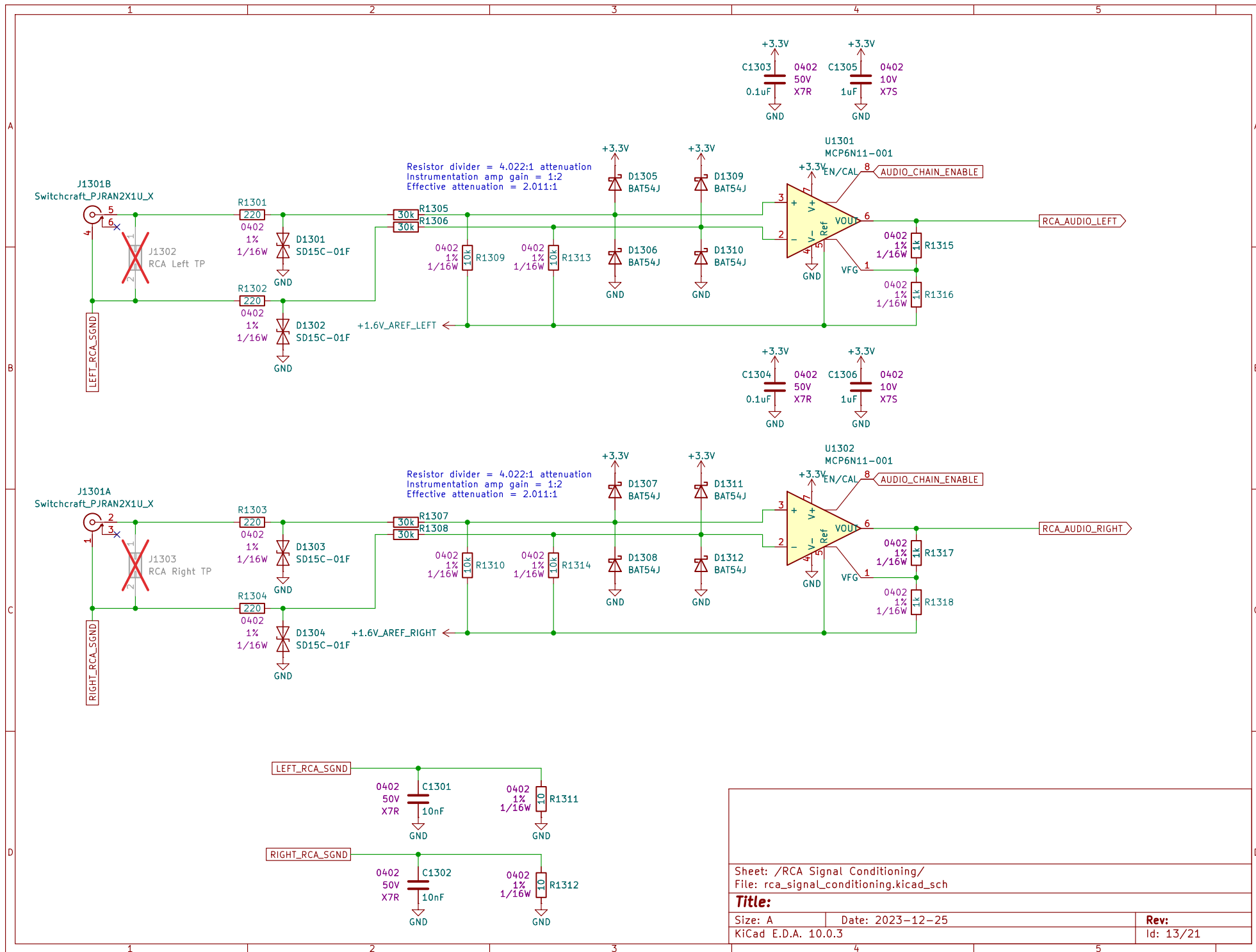
Sheet: /USB UART Bridge/
File: usb_uart_bridge.kicad_sch

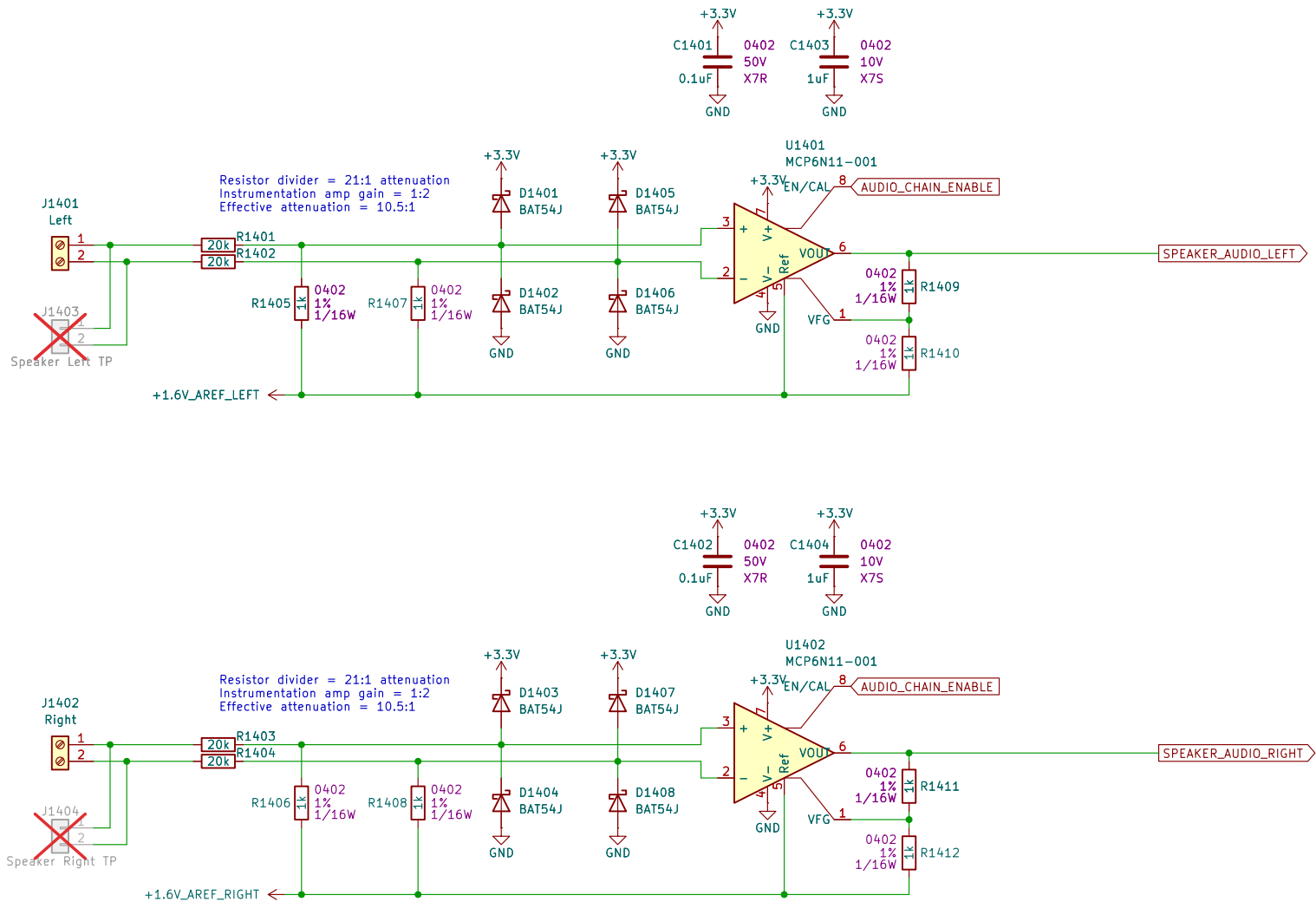
Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 9/21





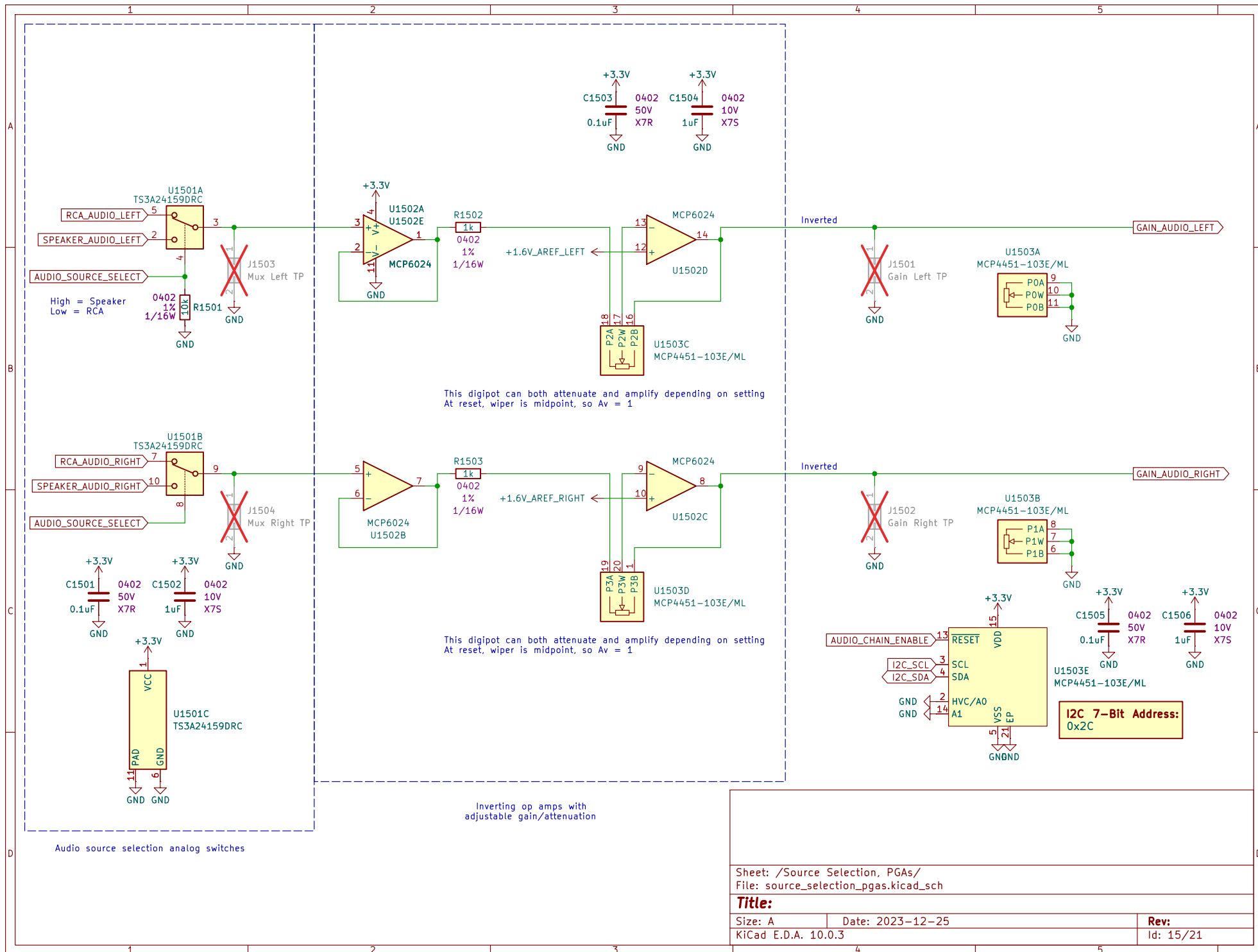


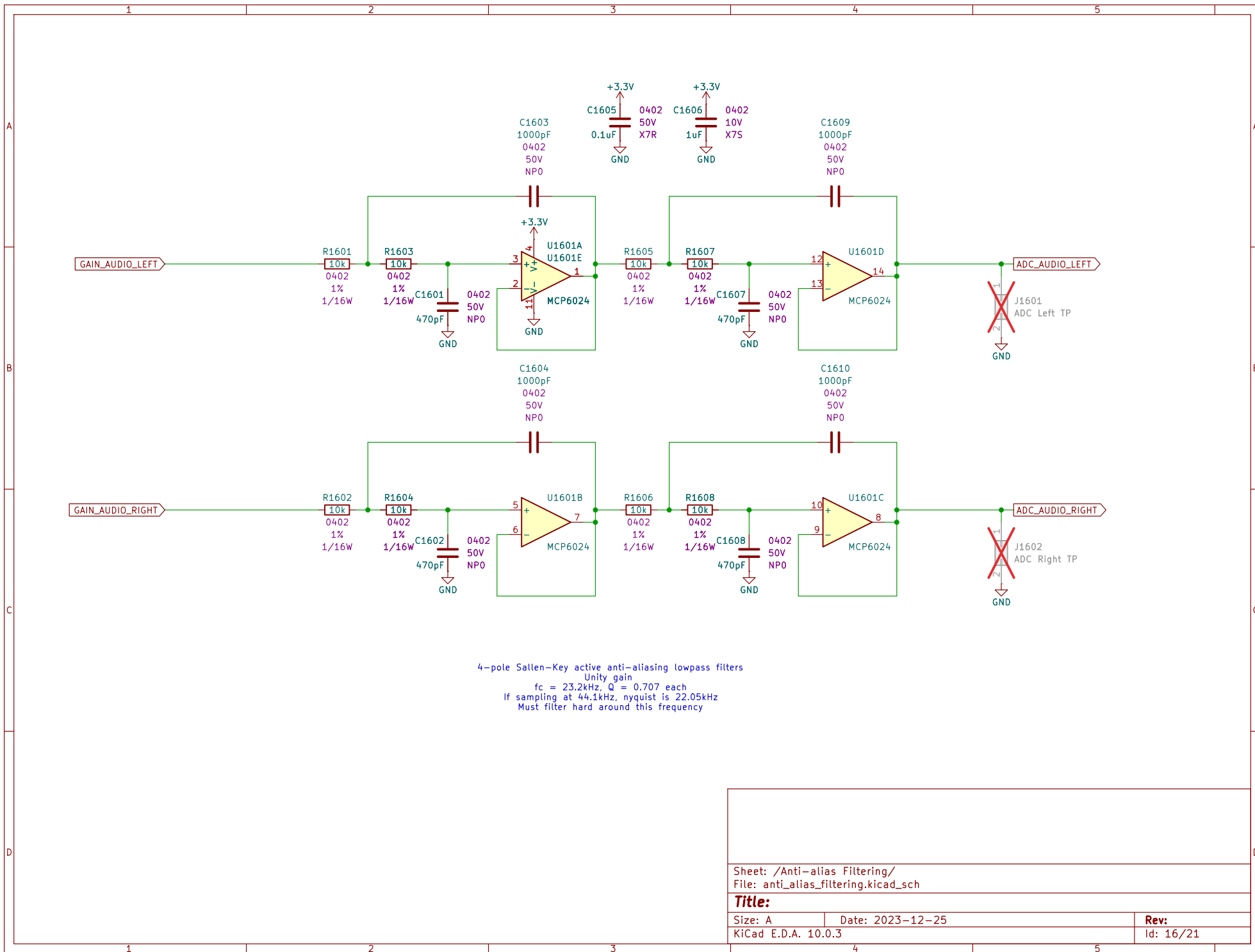
Sheet: /Speaker Signal Conditioning/
File: speaker_signal_conditioning.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 14/21





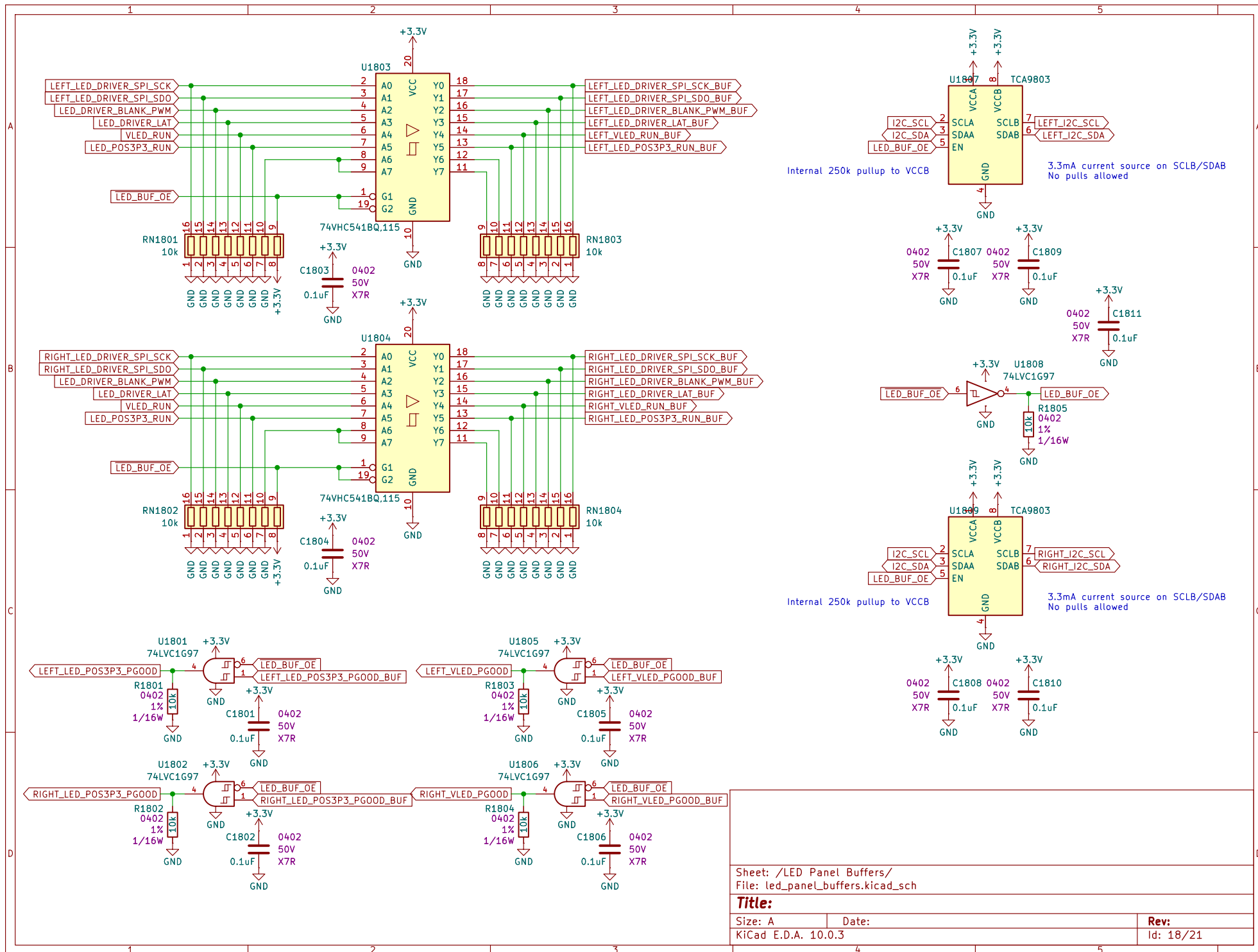
Sheet: /Anti-alias Filtering/
File: anti_alias_filtering.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 16/21





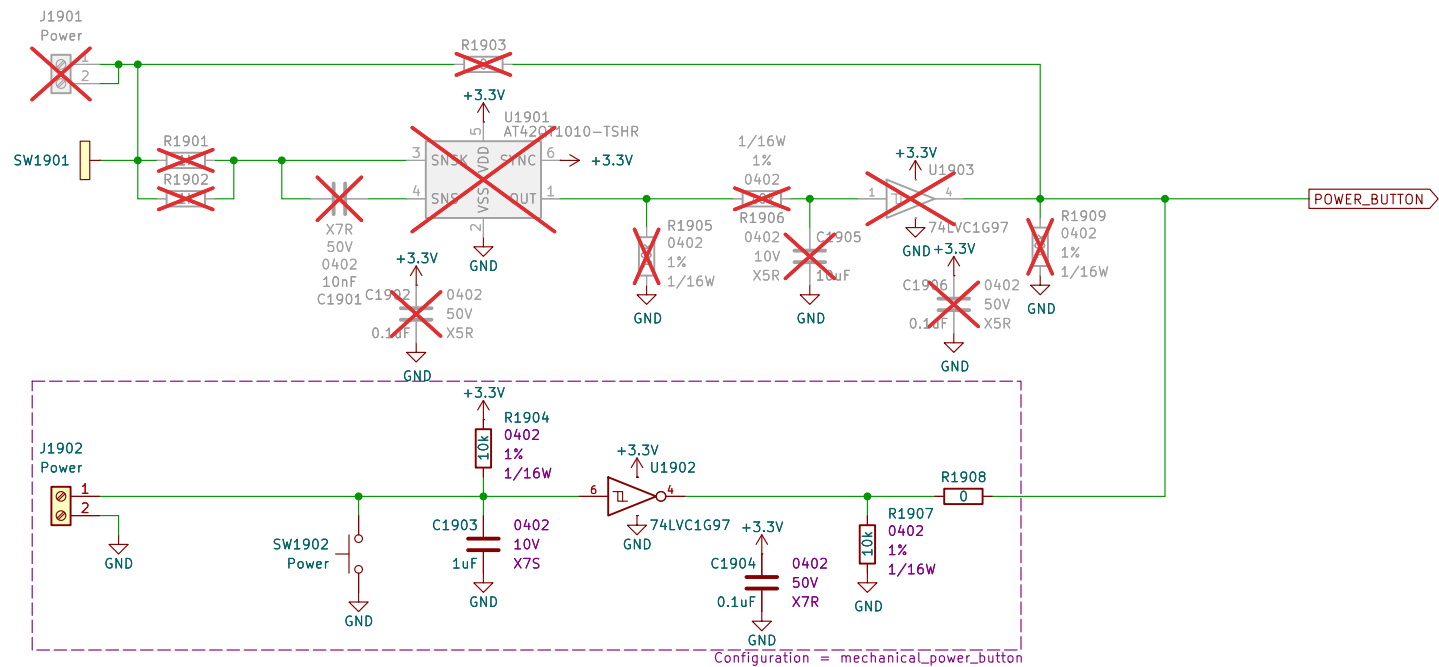
Sheet: /LED Panel Buffers/
File: led_panel_buffers.kicad_sch

Title:

Size: A
KiCad E.D.A. 10.0.3

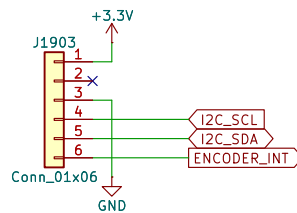
Date:

Rev:
Id: 18/21



N1901
Adafruit Rotary Encoder Module

I2C 7-Bit Address:
0x36



Sheet: /Pushbuttons/
File: pushbuttons.kicad_sch

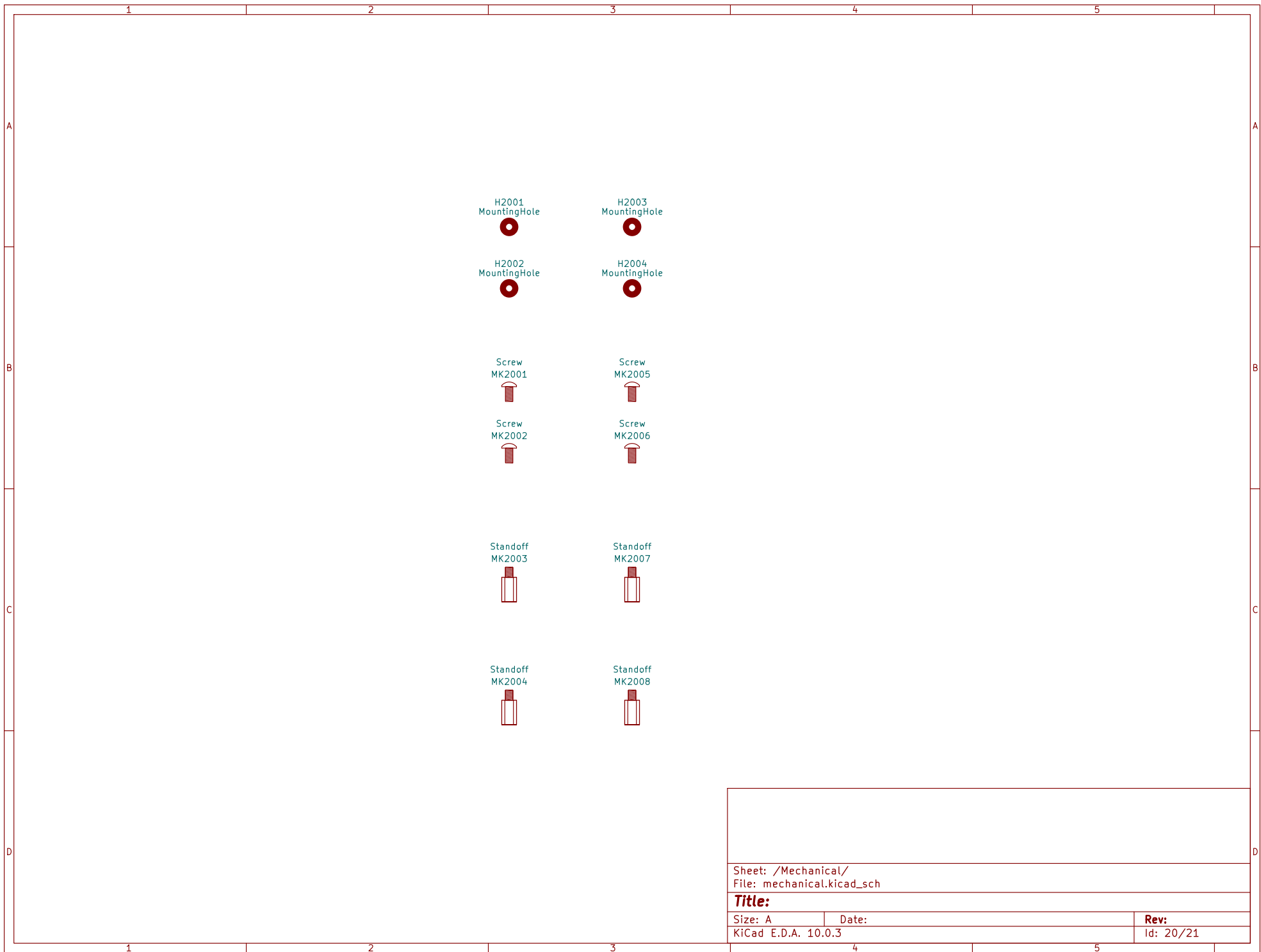
Title:

Size: A
KiCad E.D.A. 10.0.3

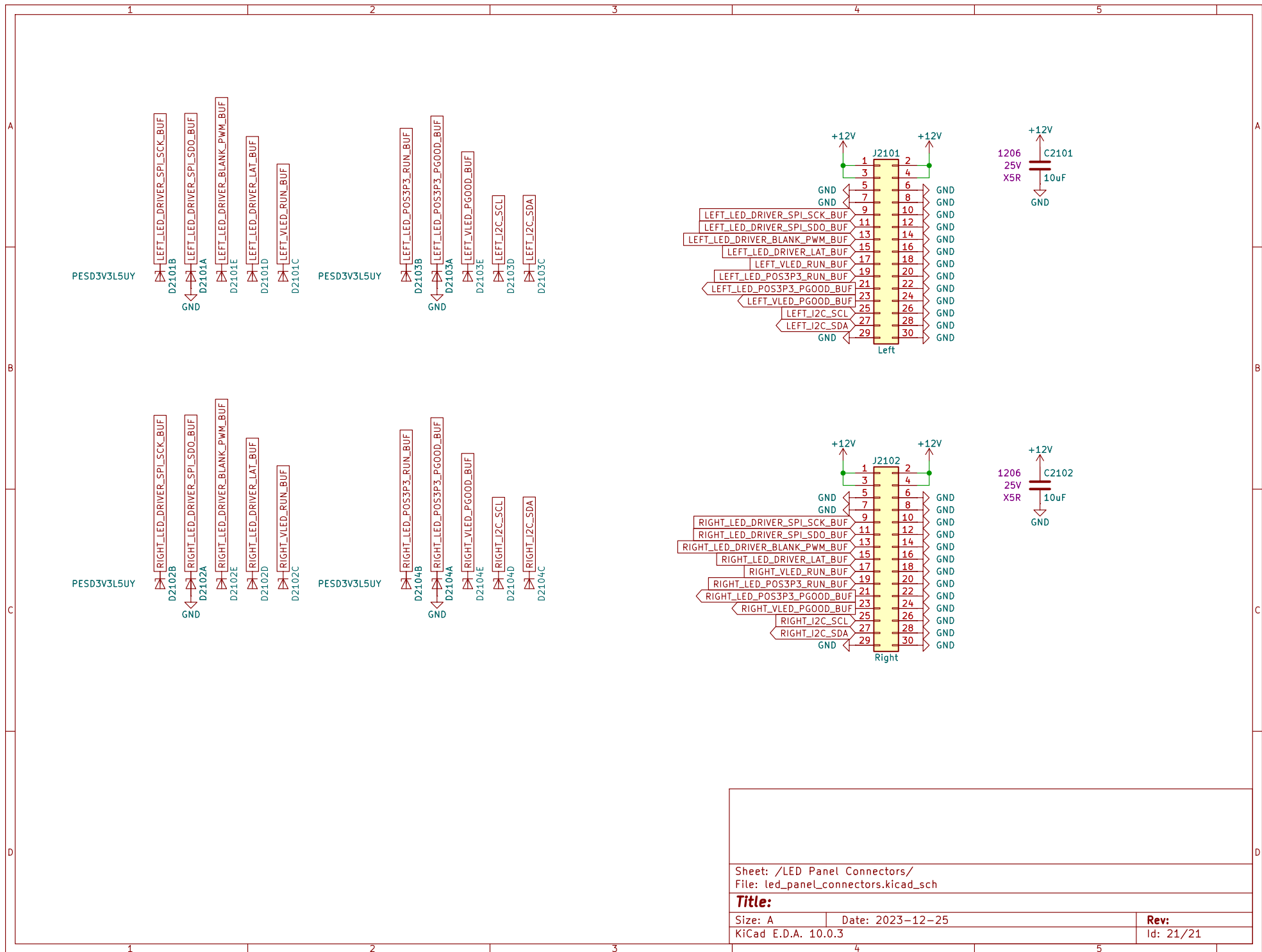
Date:

Rev:

Id: 19/21



Sheet: /Mechanical/ File: mechanical.kicad_sch		
Title:		
Size: A	Date:	Rev:
KiCad E.D.A. 10.0.3		Id: 20/21



Sheet: /LED Panel Connectors/
File: led_panel_connectors.kicad_sch

Title:

Size: A Date: 2023-12-25
KiCad E.D.A. 10.0.3

Rev:
Id: 21/21